

School of Computer Science

Research Ethics Committee (CS REC)



SOP 2.2: Taught Student AI/ML with Text Data

Standard Operating Procedure (SOP) for AI/ML with personal data, including text and social media data.

This form is a version of the CS REC ethics application form 2, for use by taught students conducting a project involving AI/ML with Text Corpora downloaded from a public repository by the student or provided by the project supervisor.

Note that there are a number of other SOPs for other types of project activity on the CS Ethics Sharepoint site. You can submit more than one ethics application for a single taught project if it has different elements that involve participants and/or personal data. If your work doesn't fit one of the SOPs then you should complete the appropriate standard ethics application form and submit it to the CS Research Ethics Committee for review.

The student:

- Must edit the yellow highlighted (variable) sections in line with the specific details of their project.
 - o Specific prompts are included in square brackets and should be replaced as appropriate, e.g. "[student name]" should be replaced with your name! (Leave it highlighted but remove the square brackets.)
- Must NOT edit any other text.
- Must read through the risk checklists (section A part 5) and confirm that the project does NOT include any of the risk factors in the list.
- Must get their project supervisor to check and confirm their changes and level of risk BEFORE starting to recruit or work with participants or personal data.

Note that you should remove any yellow highlights and italic text from Part B before sending it to data subjects (if relevant).

If your project requires you change any other text or any of the risk checklist items then you should identify these changes in section A.4 and arrange for your application to be reviewed by the CS REC. If you are unsure then ask your supervisor or someone on the CS Ethics REC for advice. (Project supervisors can no longer review non-standard applications.)

Ethics application for research where personal data is NOT obtained directly from participants.

This form should be used if the research processes personal data that are not obtained directly from the participants involved in the research. For example, if data is provided by a company or is publicly available in an online database. Note, for a dataset from a public repository or online database, only use datasets with a documented collection process (e.g. in an associated scientific paper) and the collection process should be evaluated *before* the data is used.

If you are not sure what personal data is or what processing personal data means, see *READ ME (does my research require ethics approval?)*

Sections A and B of this form must be completed and receive a favourable ethical opinion (FEO) ***before*** data is obtained and processed.

Sections A and B should be completed by the researcher. Check/Complete all sections. Edit *highlighted texts* only. Do NOT modify this form prior to ethics review.

Section B should be provided to the data subjects (the people who the data is about), **wherever possible**. It may be modified (e.g., non-applicable sub-sections deleted or formatted for publication online) after the application has been approved.

The principal investigator or supervisor should sign-off this ethics application and submit the completed form to Ps-cs-ethicsadmin@exmail.nottingham.ac.uk.

The principal investigator or supervisor is responsible for ensuring the application has been completed correctly and providing feedback to the researcher if it is required. Principal investigators or supervisors must have completed ethics training.

A data management plan (DMP) should be put in place by the researcher and be reviewed and approved by the supervisor or PI **before submission of the ethics application.¹**

If your application is a modification of an existing submission, or a new submission having only minor modifications to one previously granted a favourable ethical opinion (FEO), see Section A4.

¹ See the Data Management Plan (DMP) form and guidance.

Section A. Information to be provided to the Research Ethics Committee, along with Section B (if applicable)

1. The applicant	
Applicant's name	Yihang Cao
Applicant's role	Student 20769792
UoN email address	alyyc111@nottingham.ac.uk
Module / course details	COMP4026: Research Project in Computer Science (AI)
Supervisor/PI's name	Milena Radenkovic
Supervisor/PI's email address	milena.radenkovic@nottingham.ac.uk
Clinical advisor's name	N/A
Clinical advisor's email address	N/A

2. The project	
Project title	NLP-based Message Prioritisation for Opportunistic Routing: A Case Study of Nepal Earthquake Emergency Communication
Project content and ethical relevance	This project aims to investigate how NLP based message prioritisation can support opportunistic routing in Nepal Earthquake emergency communication. The project will use existing crisis related social media text datasets such as CrisisNLP to classify disaster messages into humanitarian categories and map them into routing priority levels. These priority levels will then be used in a simulation-based network evaluation using the ONE simulator. No new human participants will be recruited, and no interviews, surveys or field studies will be conducted. The project will only use existing research datasets for academic purposes. Any identifiers such as tweet IDs, user handles, URLs and any contact details will be removed, ignored, and not reported in the dissertation. Therefore, the project is expected to raise no more than minimal ethical risk.
Proposed start date	22/06/2026
Date and version of application	22/06/2026 Version 1
Type of application	Application of an agreed SOP
Application ID (if known) ²	CS-T-2026-SOP2.2-MR-YC-1

² An ethics application is usually allocated an ID by the CS REC *after* initial submission.

3. Change log

If this is a revision or re-submission, summarise how the application has been changed in response to the reviewers' comments, or delete if not applicable

N/A

4. Modifications and related applications

If you seek approval to modify a research project that has already received ethics approval in some way and is currently live or ongoing, or for a new research project that is very similar to one that has previously received a FEO and only contains minor modifications, please provide the application ID and describe the modifications in detail here. The reviewers will assess whether a new application is required or if this update may be appended to the original / related application. You do not need to complete the rest of this form, unless updates are also required to specific sections. If so, please list which sections have been updated and highlight them for the reviewers' attention.

N/A

5a. Research ethics checklist (part one)

Answer all questions:	Yes/No
a) Does the research pose any risks to the researchers, participants, culture, the environment, and/or the reputation of those involved in the research or the reputation of university? <i>For example...</i>	No
b) Does the data contain inappropriate images of the individuals involved?	No
c) Will the project involve inappropriate or unauthorised manipulation or representation of the data?	No
d) Could the data processing and reporting of the project outcomes cause psychological or social risks to the individuals involved?	No
e) Does the research involve travelling to countries/regions against the advice of the British Foreign Commonwealth Office.	No

5b. Data ethics checklist (part two)

Answer all questions:	Yes/No
Is there evidence, through a published paper and/or a license agreement and/or data collection ethics application with a favourable ethical opinion (FEO), that the dataset was:	
A) collected according to acceptable ethical standards?	Yes
B) collected with explicit consent from the participants?	Yes

C) collected in compliance with UK/EU data privacy laws?	Yes
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If you have answered "no" to all questions in A5a (part 1), and "yes" to all questions in A5b (part 2), you may go to Section A7.

If you have answered "yes" to any of the questions in A5a (part 1), or "no" to any of the questions in A5b (part 2), please explain below (Section A6) why this is necessary and how you plan to deal with the ethical issues raised.

6. Research involving more than minimal risk Explain here why it is necessary to conduct research involving more than minimal risk. Describe any risks posed to the researchers, participants, culture, the environment, and/or the reputation of those involved in the research or the reputation of the university, and what steps will be taken, or measures put in place to mitigate those risks. Describe what evidence there is that data has been collected ethically, legally and with consent, and any additional precautions that will be taken with the data. N/A

7. Purpose of the research a) Aims and objectives of the research Briefly describe what the research is about, its aims and objectives or what it seeks to achieve, and describe the context / motivation for the research (e.g., a thorny problem in computing, a new trend, a topical issue, the aims of a funded research project, completing module assessments, etc.). The project involves the development and evaluation of natural language processing and machine learning methods to classify crisis related social media messages into humanitarian information categories and assign routing priority levels from existing crisis related social media datasets such as CrisisNLP, with a focus on the 2015 Nepal Earthquake scenario. b) What data will be obtained and processed? Describe what data will be obtained, from whom or where, and how it will allow you to meet the aims and objectives of the research. <i>Note, if your supervisor is providing the data then they should be able to provide the necessary details for section A.7 parts b), c), d) and e) and section A.8 "data subjects".</i> The dataset consists of crisis related social media text data, including tweets or tweet like messages, tweet IDs, event information, language information and humanitarian category labels from existing research datasets such as CrisisNLP, with a focus on the 2015 Nepal Earthquake scenario. The data will be processed using Python, Pandas, scikit-learn and NLP models such as TF-IDF with Logistic Regression and BERT. c) How will the data be obtained? Describe how the data will be obtained. Note that web crawling and screen scraping may violate intellectual property rights, terms and conditions, and user privacy.³ You must be sure, and provide evidence with this ethics application, that any data collected in these ways is in the public domain and that you process the data in ways the data subjects (the people identified in or by the data) would expect or find fair.⁴

³ <https://parissmith.co.uk/blog/web-crawling-screen-scraping-legal-position/>

⁴ <https://ico.org.uk/about-the-ico/news-and-events/news-and-blogs/2022/05/ico-fines-facial-recognition-database-company-clearview-ai-inc>

The Applicant(student) has obtained the data from ethically existing public research datasets such as the CrisisNLP.

d) Permission to use the data

Provide evidence the data will be or was lawfully obtained and that you are entitled to use it (e.g., online license, UoN contract reference, headed letter; you may submit supporting documentation along with this application). You cannot use special category data – inc. race or ethnicity, sexual orientation or sexual activity, political opinions, religious or philosophical beliefs, biometric identifiers, genetic or health data – without the explicit consent of the data subject.⁵ You must provide evidence that explicit consent has been given if you wish to use such data.

Identify whichever option applies:

Yes: Data from a third form a third party or public repository:

The Applicant has obtained permission to use the data by agreeing to the Terms of Use of the CrisisNLP related social media dataset with the Crisis Computing team at Qatar Computing Research Institute (QCRI). The Terms of Use state that the data was collected for humanitarian computing research using publicly available APIs. The project will use the data only for academic research on crisis message classification and emergency communication, will not redistribute the raw dataset, will keep the dataset contents confidential, will delete the dataset after completion if required and will cite the corresponding paper in the dissertation. A screenshot of the Terms of Use will be attached as supporting documentation.



Terms of Use

The Crisis Computing team at Qatar Computing Research Institute (QCRI) collects data for research on humanitarian computing. This is publicly-available data collected using publicly-available application programming interfaces (APIs).

IN NO EVENT SHALL QCRI OR QCRI'S AFFILIATES BE LIABLE TO THE REQUESTER OR TO THE USER OF THE DATA SUPPLIED FOR ANY SPECIAL, INCIDENTAL, PUNITIVE, INDIRECT, EXEMPLARY, OR CONSEQUENTIAL DAMAGES. THIS DATA IS PROVIDED ON AN "AS IS" BASIS AND YOUR USE OF IT IS AT YOUR SOLE RISK.

To the extent permitted by applicable law, you agree to defend, indemnify and hold harmless QCRI, and its parent QF, officers, directors, employees and agents, from and against any and all claims, damages, obligations, losses, liabilities, costs or debt, and expenses (including but not limited to reasonable attorney's fees) arising from: (i) your use of and access to this data; (ii) your violation of any term of these terms in this Agreement; (iii) your violation of any third party right, including without limitation any copyright, property, or privacy right; or (iv) any claim that your Content caused damage to a third party. This defense and indemnification obligation will survive this Agreement and your use of the data.

By downloading or using any of the material available in this dataset, you automatically acknowledge to the following conditions:

- You will not defame, abuse, harass, stalk, threaten or otherwise violate the legal rights (such as rights of privacy and publicity) of others;
- You will not publish, post, distribute or disseminate any defamatory, infringing, obscene, indecent or unlawful material or information;
- You will use the data only for research on humanitarian computing;
- You will keep the contents of these dataset(s) confidential (you can always share the tweet-ids);
- The dataset(s) will be deleted by you upon our request at any time (you can always keep the tweet-ids);
- The dataset(s) will be deleted by you upon completion of the research (you can always keep the tweet-ids);
- You will cite the corresponding paper in derived publications.

No: Data previously collected by the Applicant/Project Supervisor:

The data collection was given a favourable ethical opinion (FEO) by the [specify research ethics review board e.g., Computer Science Research Ethics Committee] as [ethics committee application/approval ID] ([specify title of the research project]). The data processing for this project will adhere to terms under which it was collected. [Attach data collection ethics documentation e.g., participant information sheet, ethics checklist, consent forms etc]

e) Constraints on use of the data

Specify on constraints placed on use of the data by the party or parties it was obtained from, if applicable

⁵ <https://ico.org.uk/for-organisations/guide-to-data-protection/guide-to-the-general-data-protection-regulation-gdpr/lawful-basis-for-processing/special-category-data>

Identify all that apply:

Yes: The tweet text, tweet IDs, user handles, URLs and any other potentially identifying information must not be published in the dissertation or project outputs in a way that could identify individuals. The project will only report anonymised examples where necessary and will mainly present aggregate classification and simulation results.

Yes: The dataset must not be republished, redistributed or uploaded online. The dataset contents will be kept confidential and used only for this academic research project. The dataset will be deleted after completion of the research if required by the dataset Terms of Use.

8. Data processing

Data subjects

Specify the categories of people the data concerns or is about (e.g., students, household members, members of staff at X company, members of the public, Twitter users, etc.)

Twitter users or social media users whose crisis related posts are included in the existing CrisisNLP datasets. The project focuses on crisis related messages associated with the 2015 Nepal Earthquake. No new participants will be recruited and the applicant will not contact any data subjects directly.

Pre-processing of data

i. Pre-processing procedure(s)

Specify any processing that will be done to the data before it is analysed (e.g., annotation, labelling, cleaning, etc.).

The data will be pre-processed using a standard natural language processing pipeline: the data will be pre-processed using a standard NLP pipeline: removing or replacing URLs, user mentions, account handles, duplicated messages, missing values, excessive whitespace and other noisy text. Normalising text where appropriate and converting the cleaned text into numerical features for classification.

ii. Anonymisation / pseudonymisation applied

Specify how pre-processed data will be anonymised or pseudonymised if applicable (e.g., pseudonyms or alphanumerical IDs will be used instead of real names, identifying information will be removed from free text responses, etc.)

Identify whichever applies:

No: The data is already pseudonymised, i.e. subjects are identified by a dataset-specific participant ID.

Yes: The data will be pseudonymised using the following procedure by removing tweet IDs, account handles, URLs and any direct contact details from the working dataset and by using internal non-identifying message IDs for analysis.

Identify all that apply:

	No: [Data: specify data type(s), e.g., textual data and metadata] is already anonymised (i.e. participants are not re-identifiable). Yes: The textual data and metadata will be anonymised using the following procedure by removing or replacing account handles, URLs, tweet IDs and any direct contact details. No: [Data: specify data type, e.g. image, audio] cannot be anonymized before analysis because [specify reasons]. [Specify any restrictions on the publication of identifying information in reports if applicable e.g., images will not be published in reports except for participants who consented to this]
Data analysis	<i>Specify the analytic method(s), procedure(s), protocol(s) or technique(s) that will be used to process and analyse the data. Insert a URL to an example of each (or similar where a new method, procedure, protocol or technique is used).⁶</i> The crisis related social media text data will be analysed using natural language processing and machine learning methods, including TF-IDF with Logistic Regression as a baseline classifier and a transformer based model such as BERT.
Data sharing	<i>Specify any third-party recipients or services the data is shared with and how they are GDPR compliant and protect the data</i> N/A
Data storage	<i>Specify the data asset's name, location and owner.</i> Dataset name: NLP-based Message Prioritisation for Opportunistic Routing: A Case Study of Nepal Earthquake Emergency Communication Dataset location: University of Nottingham OneDrive Dataset owner: Yihang Cao

9. Provision of information to data subjects

Data protection law requires that when data is NOT obtained directly from data subjects they be provided with the information in Section B within 1 month of the data being obtained, unless:

- The individuals concerned already have the information
- Providing the information to the individuals concerned would be impossible
- It would involve a disproportionate effort to provide the information to individuals

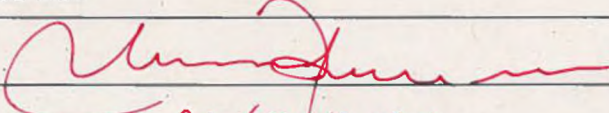
⁶ See, for example, [https:// ... insert link to example of method, procedure, protocol, technique](#)

Do the individuals concerned already have the information in Section B (or equivalent)?	No
<i>If "Yes", please specify how they were provided with this information below. The following items are then Not Applicable.</i> [If yes, explain how, e.g. that the current project falls within the range of use described to participants when the data was collected]	
If the individuals concerned do NOT have the information in Section B (or equivalent), is it possible to provide them with it?	No
If the individuals concerned do NOT have the information in Section B (or equivalent) but it is possible to provide them with it, is the effort involved proportionate to the research? <i>If "Yes" then you must ensure that the information in Section B is provided to the individuals concerned.</i>	NA
If the individuals concerned do NOT have the information in Section B (or equivalent), and it is not possible or proportionate to the research to provide it, state why: Identify whichever applies: No: The individuals concerned already have the information in Section B (or equivalent) Yes: It is not possible to provide this information to individuals in the dataset because contact details not available. Yes: It is not proportionate to provide this information to individuals in the dataset because the project is low risk and time constraints.	

10. Further information <i>Please provide any further information which may affect consideration of the ethics involved in the proposed research, e.g., the nature and sensitivity of the data, unusual circumstances, use of AI components, software analytics or biometric software used to process data, etc., including any services not made available by the University and their data use and retention policies. Please include any supporting documentation with this application.</i> The NLP and machine learning algorithms including TF-IDF with Logistic Regression and BERT based text classification will be implemented on the applicant's password protected workstation and University provided computing resources. Data processing will be carried out using Python, Pandas, scikit-learn and possibly PyTorch or Hugging Face Transformers. Network simulation will be conducted using the ONE simulator. Identifiable data will not be uploaded to public repositories or shared with non-University online services. The project will not use biometric software and will not make automated decisions that have legal or significant effects on individuals.

11. Applicant declaration	
Please confirm each of the following statements:	Yes/No
I confirm that I've completed all variable sections in Parts A and B (indicated by the highlighted text) to reflect my project	Yes
I confirm that I have not changed any of the fixed sections	Yes

I confirm that the data does not contain inappropriate images of the individuals involved.	Yes
I confirm that the project will not involve inappropriate or unauthorised manipulation or representation of the audio/video data.	Yes
I confirm that the data processing and reporting of the project outcomes will not cause psychological or social risks to the individuals involved.	Yes
I confirm that this work will NOT involve making automated decisions that have legal effect or significantly impact participants.	Yes
The research is minimal risk (see section A.5)	Yes
The data has been obtained lawfully	Yes
The data management plan (DMP) form has been completed	Yes
I recognise that it is my responsibility to check whether a DPIA is necessary and can confirm that it is necessary/unnecessary (link to DPIA here). [If you need further information, please contact the DPO.]	Unnecessary
I confirm that I will conduct the work in accordance with the description here.	Yes
Applicant	Yihang Cao
Date	22/06/2026

12. Supervisor/PI declaration	
Please confirm each of the following statements:	Yes/No
I have completed ethics training	Yes
The proposed research complies with the UoN Code of Research Conduct and Research Ethics	Yes
I have reviewed and approve the DMP <i>The supervisor/PI must refer the DMP to CS REC for approval if the answer is NO to Sections 6.2, 7.4, 7.5 or 8.2. If the applicant and the PI are the same, the DMP should be submitted with this application for review by CS REC.</i>	Yes
I confirm that the research is minimal risk and that the application complies with the SOP	Yes
Supervisor/PI	
Date	22/06/2026

Section B. Information to be provided to data subject

Note: see section A9 to confirm whether you will provide this information to data subjects.

Notice of personal data processing for research purposes

The School of Computer Science, University of Nottingham has obtained personal data from **enter source** for use in research.

Data obtained

The Applicant(student) has obtained the data from ethically existing public research datasets such as the CrisisNLP.

Aims and objectives of the research

The project involves the development and evaluation of natural language processing and machine learning methods to classify crisis related social media messages into humanitarian information categories and assign routing priority levels from existing crisis related social media datasets such as CrisisNLP, with a focus on the 2015 Nepal Earthquake scenario.

Purpose(s) of processing

Data collected during the research that identifies you may be:

- Analysed to meet the aims and objectives described above
- Reviewed and discussed in supervision sessions between researchers and their supervisors or in research meetings between members of the research team, including project partners
- Used in scientific works, including presentations, reports and publications stored in databases and posted online and in marketing materials that promote the research and its findings

Automated decision-making

Data will not be used for automated decision making, i.e. wholly automated decisions that have legal effect or significantly impact participants.

Lawful basis of processing

We collect personal data under the terms of the University of Nottingham's Royal Charter and in our capacity as a teaching and research body to advance education and learning. We thus process your data on the legal basis that our research is in the public interest, we have legitimate interests and / or that you consent to data processing in freely and voluntarily participating in our research activities.

Retention of your data

Your data may be archived for future use for public interest, scientific or historical research purposes or statistical purposes, subject to the implementation of technical and organisational measures that safeguard your data, your legal rights and your freedoms.

Disclosure of your data to another party

Data will not be shared with another party.

Ethics approval

The use of your data for the purposes specified above follow a standard protocol agreed by the School of Computer Science Research Ethics Committee (CS REC), SOP2.2 and has been reviewed by my project supervisor, **Milena Radenkovic**.

You can contact the CS REC by email at Ps-cs-ethicsadmin@exmail.nottingham.ac.uk

Our contact details

School of Computer Science
University of Nottingham
Jubilee Campus
Wollaton Road
Nottingham NG8 1BB
United Kingdom

Our Data Protection Officer

Data Protection Officer
Legal Services
A5 Trent Building
University of Nottingham
University Park
Nottingham NG7 2RD
United Kingdom

Your rights

You have the legal right:

- To be informed about the collection and use of personal data (as per this notice).
- To access and receive a copy of your personal data, and other supplementary information, on request.
- To object to and restrict data processing if you think we are not complying with data protection law, and to rectify inaccuracies.
- To be forgotten, i.e., to have your personal data erased.
- To data portability and to obtain your data in an accessible and machine-readable format if appropriate, or to transfer your data to another organisation if technically feasible.
- To complain to the Information Commissioner's Office: <https://ico.org.uk/make-a-complaint>